

$$w_{22}^{\text{new}} = 0.5928 - 0.000636 = 0.5922$$

## II Iteration

$$y_1 = (x_1 \cdot \Delta w_{11}) + (x_2 \cdot \Delta w_{21}) = (0.85 \cdot 0.099) + (0.9 + 0.7973) = 0.03965 + 0.75752 = 0.75237$$

$$f(y_1) = \frac{1}{1 + e^{-0.25237}} = 0.549552$$

$$y_2 = (x_1 \cdot \Delta w_{12}) + (x_2 \cdot \Delta w_{22}) = (0.36 \cdot 0.396882) + (0.9 - 0.591889) = 0.138887 + 0.5327001 = 0.571587$$

$$f(y_2) = 0.553855$$

$$y_3 = (h_1 \cdot \Delta w_{13}) + (h_2 \cdot \Delta w_{23}) = (0.75237 \cdot 0.2094) + (0.0715 \cdot 0.8704) = 0.20267 + 0.58956 = 0.78723$$

$$f(y_3) = 0.58723$$

$$\sum \text{error}_2: 0.5 - 0.887232 = -0.38723$$

Back. prop. 12)

$$\delta_3 = y_3(1 - y_3)(y_1 - y_3) = 0.8872(1 - 0.6872)(0.5 - 0.6872) = 0.6872 \cdot 0.3128 \cdot (-0.1872) = -0.04024$$



$$\delta_2 = y_2 (y - y_2) (\Delta w_{23} - \delta_3) = 0.6519 (1 - 0.6619) \cdot 0.8704 \cdot (-0.04024) = -0.00284$$

$$\delta_5 = y_5 (1 - y_5) (\Delta w_{13} - \delta_3) = 0.629812 \cdot (1 - 0.629622) \cdot 0.02694 \cdot (-0.04024) = 0.00235$$

$$U_{23}' = 0.1 \cdot \delta_3 \cdot y_2 + \Delta w_{23} = (0.1 \cdot (1 - 0.0402) \cdot 0.6619) + 0.8704 = 0.86723$$

$$\Delta U_{13}' = 0.1 \cdot \delta_3 \cdot y_1 + \Delta w_{13} = 0.1 \cdot (1 - 0.0402) \cdot 0.63582 + 0.2594 = 0.2662$$

$$\Delta w_{11}' = 0.1 \cdot \delta_1 \cdot x_1 + \Delta w_{11} = (0.1 \cdot (1 - 0.00238) \cdot 0.35) + 0.099056 = 0.09893$$

$$\Delta w_{21}' = 0.1 \cdot \delta_1 \cdot x_2 + \Delta w_{21} = (0.1 \cdot (1 - 0.06238) \cdot 0.9) + 0.797380 = 0.7972$$

$$U_{22}' = 0.1 \cdot \delta_2 \cdot x_2 + \Delta w_{22} = (0.1 \cdot (1 - 0.00784) \cdot 0.9) + 0.591889 = 0.5848$$

$$\Delta w_{12}' = 0.1 \cdot \delta_2 \cdot x_1 + \Delta w_{12} = (0.1 \cdot (1 - 0.00784) \cdot 0.35) + 0.39682 = 0.3965$$

III Iteration.

$$y_5 = (x_1 \cdot \Delta w_{11}') + (x_2 \cdot \Delta w_{21}') = (0.35 - 0.09893) + (0.9 \cdot 0.7972) = 0.0345255 + 0.71748 = 0.7521$$

$$f(y_5) = \frac{1}{1 + e^{-0.722}} \approx 0.6285$$



$$y_2 = (x_1 \cdot \Delta W_{12}) + (x_2 \cdot \Delta W_{22}) = 10,35 \cdot 0,3985) + (0,9 \cdot 0,5848) = 0,138875 + 0,52832 = 0,667195$$

$$f(y_2) = \frac{1}{1 + e^{-0,5001}} \approx 0,6604$$

$$y_3 = (h_1 \cdot \Delta W_{13}) + (h_2 \cdot \Delta W_{23}) = (0,7521 \cdot 0,2654) + (0,651 \cdot 0,86775) = 0,2001338 + 0,577127 = 0,77726$$

$$f(y_3) = \frac{1}{1 + e^{-0,77726}} \approx 0,6851$$

$$\text{Error}_3: 0,15 - 0,6851 = -0,5351$$